

# AIDAN M. MOKALLA

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## EDUCATION

GPA: 3.6/4

**Reed College**, Portland, Oregon, USA,

2025

+ Spanish minor, sole member of graduating class with double majors: Computer Science & Linguistics,

**Bryn Mawr College**, Dushanbe, Tajikistan,

2024

+ Fellowship funded by international non-profit *American Councils* to research Persian language varieties,

**La Universidad de Carlos III** and **Middlebury College**, Madrid, Spain,

2023

+ Spanish-language courses including International IP Law, Data Science, and Reinforcement Learning.

## PUBLICATION

**Interpretable Signals Reveal Failure-Relevant Representations in VLA Models,**

ICML 2026

+ Discovered representational mechanistic predictors of vision-language-action models task failure,

**Fault- and Network-Agnostic Agreement,** *with* Erica Blum,

TCC 2026

+ Proved tight running-time bounds for all algorithms achieving network-agnostic Byzantine consensus,

**Language-Agnostic Concept Spaces Found in Large Language Models,** *thesis, with* Greg Anderson 2025

+ Designed, implemented, and evaluated novel architecture to interpret transformer-based LLMs by leveraging natural-language translation of ground-truth semantics. My approach learns convergent “concept spaces” over LLM activations in parallel, yielding more meaningful LLM interpretations than prior work,

**Modulated Signal Filtering: Wouldn't it be Noise?,** *thesis, with* Sameer ud Dowla Khan,

2025

+ Collected over 3,000 participant responses to investigate how listeners use unique cognitive strategies to parse spoken speech versus song lyrics using deep autoregressive analytic methods,

+ Demonstrated how we use contextually-differentiated mechanisms to filter noise from the linguistic signal by making significantly better predictions of how people interpret unclear speech within the *Noisy Channel Model*.

## LEADERSHIP & EXPERIENCE

**AlgoVerse AI Research,** *Machine Learning Researcher & Mentor*

September 2025 – present

+ Investigating and publishing methods of explainable AI and climate science, mentoring undergraduate researchers,

**Linguistic Diversity Initiative,** *Lead Volunteer Ambassador,*

June 2023 – August 2025

+ Leading a team of three translators, I compiled and published dictionaries of terminology used in Portland Public's science curricula, translated into 6 languages, for households with low English proficiency, already enabling over 800 families to engage in their students' education,

+ Taught seminars to students and faculty about sociolinguistic research on language use in undergraduate settings,

+ Organized and lead twice-monthly meetings of over 50 volunteers and larger events,

**Reed College Student Body,** *Senator & Honor Councillor,*

September 2021 – May 2025

+ Peer-elected to serve in faculty committees including the *Committee on Academic Policies and Planning* and *Committee on Computer Policy and Planning*, disburse 6-figure Student Body Fund and other resources to college organizations, and interview and hire all student applicants for on-campus roles,

+ Mediated and resolved conflicts between students, professors, and other members of the college community,

**Stoel Rives, LLC,** *Intern Practise Assistant,*

May – August 2022

+ Implemented compliant automation of docketing, court appearances, and other events from firm-wide databases,

+ Maintained communication on behalf of attorneys while drafting and proofreading attorney correspondences.

## SKILLS & ACCOLADES

**Computational:** Python, various data and NLP pipelines, PyTorch, NumPy, JS, Next.js, C++, PHP, SQL, Swift,

**Awards:** Presidential Commendation for Academic Excellence, Reed College Music Scholarship, National Diversity in STEM Grant, Department of State Critical Languages Scholarship,

**Certifications:** Federally eligible for non-competitive civil service, licensed in Conflict Mediation & Resolution, CPR,

**Languages:** Fluent Spanish, English, intermediate Farsi, Tajik, French, basic Portuguese, Dari, Chinese (Mandarin).